Monte Carlo and Quasi-Monte Carlo Integration

Pricing Discretely Monitored Arithmetic Asian Call Options in the Black–Scholes Model

Bischoy Bert

06 November 2026

Results

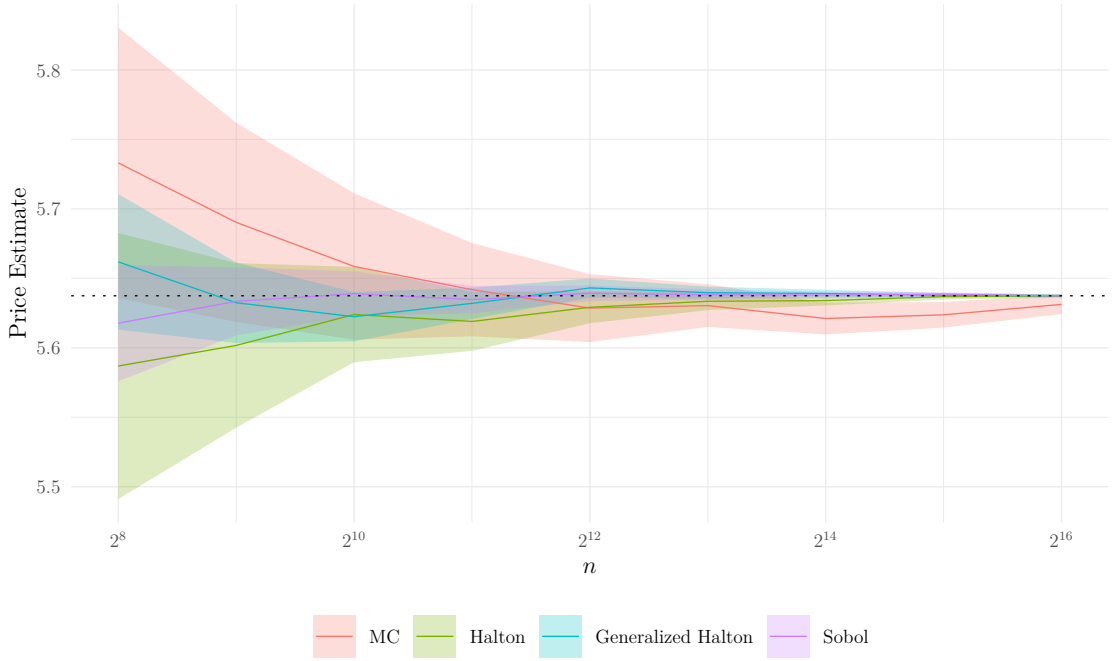


Figure 1: Geometric Asian call at $d = 52$ with the sequential construction. Estimates include 95% confidence bands. Dotted: exact price.

Estimator	Coverage	Intervals
MC	0.975	81
Halton	0.963	81
Generalized Halton	0.975	81
Sobol	0.926	81

Table 1: Share of the confidence intervals for the geometric call that contain the exact price, across all d and both constructions.

d	Geometric (exact)	Arithmetic, MC	Arithmetic, RQMC
1	10.450584	10.457631 ± 0.011126	10.450686 ± 0.000139
4	6.733487	6.939785 ± 0.006723	6.939554 ± 0.000170
12	5.940200	6.153189 ± 0.006830	6.156058 ± 0.000182
52	5.637432	5.847440 ± 0.007208	5.853817 ± 0.000210
252	5.565509	5.782412 ± 0.005677	5.781623 ± 0.000182

Table 2: Asian call prices from nM evaluations ($n = 2^{16}$, $M = 100$) $\pm$ the half-width 95% confidence band. MC uses the sequential construction, RQMC the Sobol sequence with the Brownian bridge.

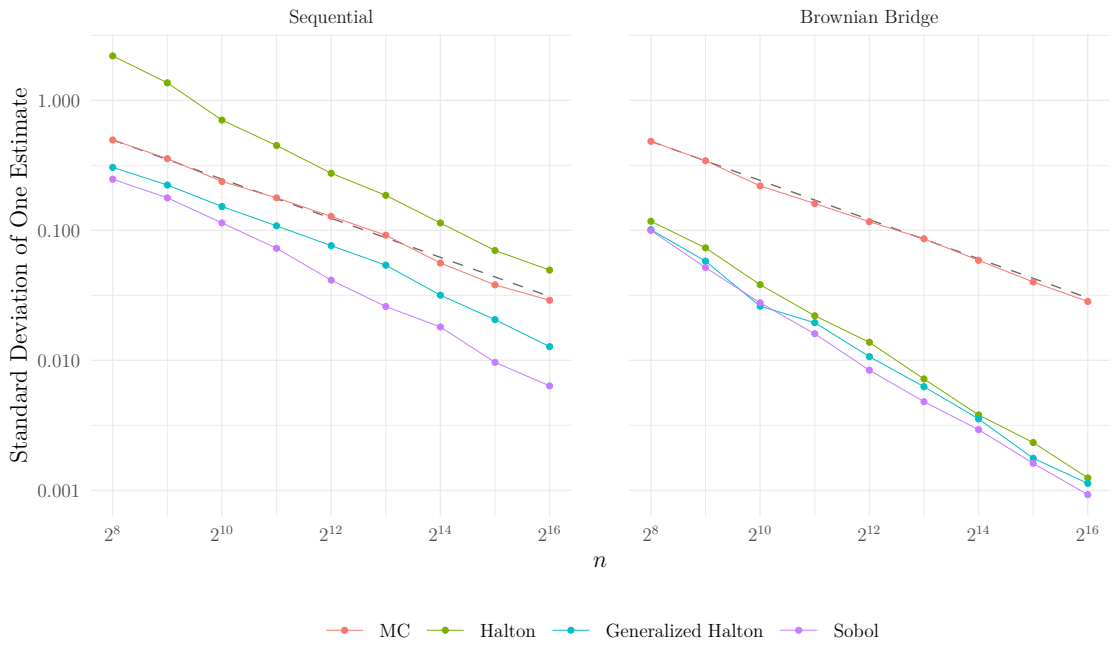


Figure 2: Standard deviation (RMSE) of a single n-point estimate of the arithmetic call at $d = 252$. Dashed: slope $-\frac{1}{2}$ anchored at the MC value at $n = 2^8$.

	$d = 1$	$d = 4$	$d = 12$	$d = 52$	$d = 252$
Sequential					
MC	0.49	0.52	0.48	0.49	0.52
Halton	0.93	0.82	0.73	0.80	0.69
Generalized Halton	0.93	0.84	0.72	0.71	0.57
Sobol	0.91	0.86	0.76	0.69	0.68
Brownian Bridge					
MC	0.49	0.52	0.49	0.50	0.51
Halton	0.93	0.85	0.80	0.81	0.82
Generalized Halton	0.93	0.85	0.81	0.84	0.81
Sobol	0.91	0.88	0.84	0.84	0.84

Table 3: Empirical convergence: negative slope of a least-squares fit of $\log \hat{\sigma}_n$ on $\log n$.

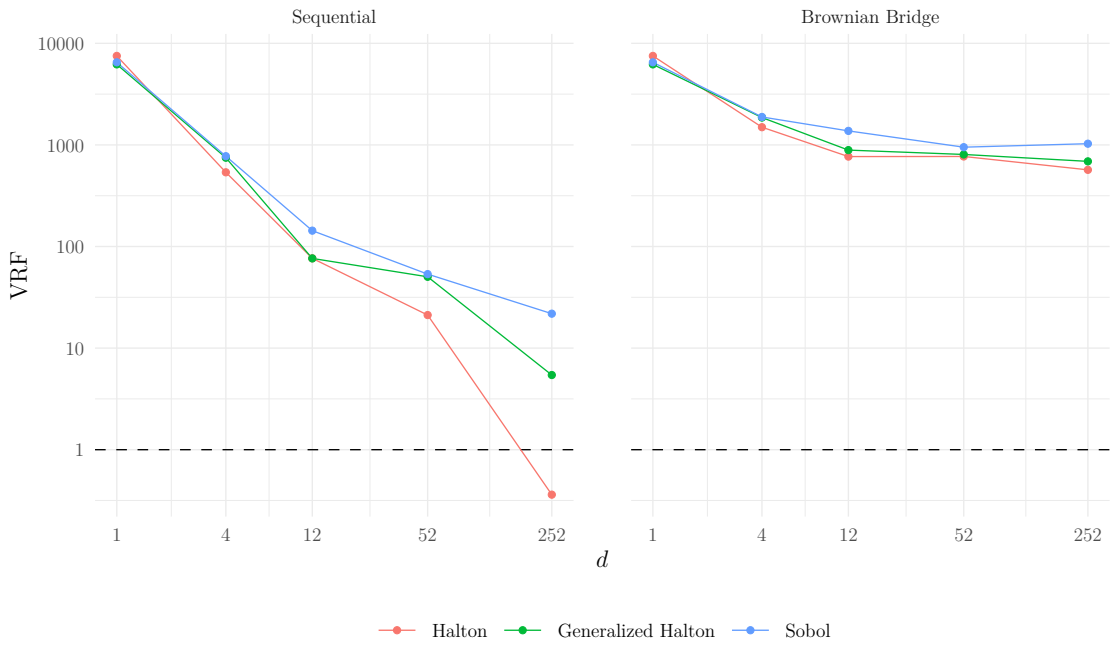


Figure 3: Variance reduction factor of RQMC over MC at $n = 2^{16}$ (log scales). Below the dashed line, RQMC is less efficient than MC.

Construction	Point Set	$d = 1$	$d = 4$	$d = 12$	$d = 52$	$d = 252$
Sequential	Halton	7,500	540	76	21	0.36
Sequential	Generalized Halton	6,200	750	76	50	5.4
Sequential	Sobol	6,500	770	140	53	22
Brownian Bridge	Halton	7,500	1,500	770	770	570
Brownian Bridge	Generalized Halton	6,200	1,900	890	810	690
Brownian Bridge	Sobol	6,500	1,900	1,400	950	1,000

Table 4: VRFs at $n = 2^{16}$, rounded to two significant digits.

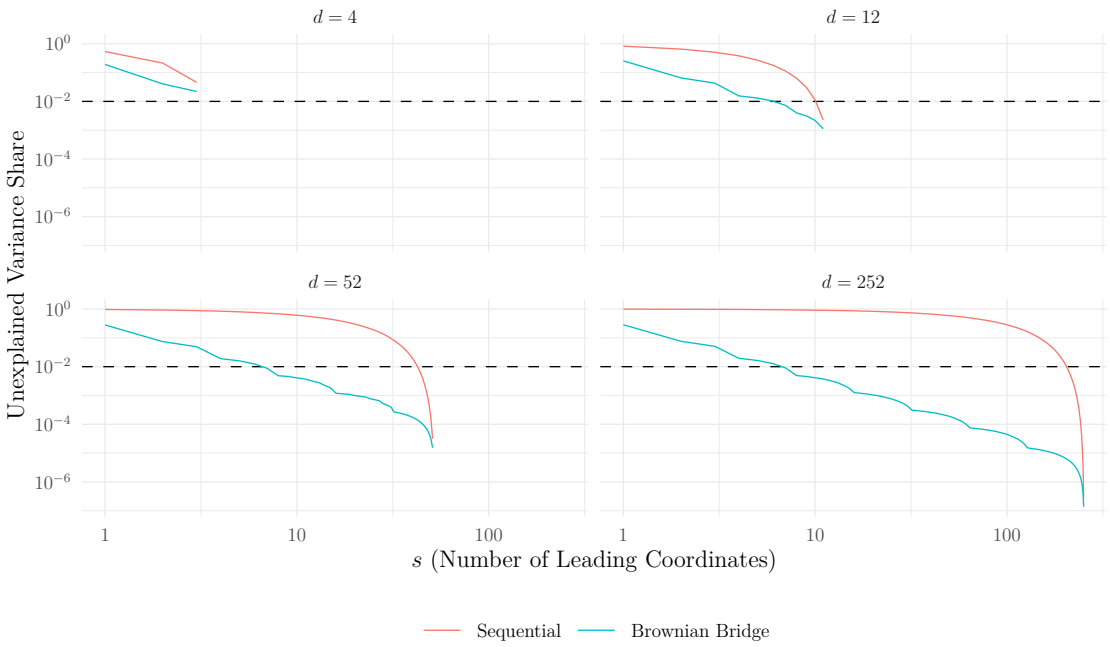


Figure 4: Share of the variance of the arithmetic payoff not explained by the first s coordinates. The dashed line marks the 1% level that defines the truncation dimension d_t.

Construction	$d = 1$	$d = 4$	$d = 12$	$d = 52$	$d = 252$
Sequential	1	4	11	43	205
Brownian Bridge	1	4	7	7	7

Table 5: Truncation dimension d_t of the arithmetic payoff.